JavaScript is disabled on your browser.

compact1, compact2, compact3

java.util.stream

Enum Collector.Characteristics

java.lang.Object
    java.lang.Enum<Collector.Characteristics>
        java.util.stream.Collector.Characteristics

All Implemented Interfaces:

`Serializable, Comparable<Collector.Characteristics>`

Enclosing interface:

`Collector<T,A,R>`

```
public static enum Collector.Characteristics
extends Enum<Collector.Characteristics>
```

Characteristics indicating properties of a `Collector`, which can be used to optimize reduction implementations.

Enum Constant Summary

Enum Constants

Enum Constant	Description
CONCURRENT	Indicates that this collector is *concurrent,* meaning that the result container can support the accumulator function being called concurrently with the same result container from multiple threads.
IDENTITY_FINISH	Indicates that the finisher function is the identity function and can be elided.
UNORDERED	Indicates that the collection operation does not commit to preserving the encounter order of input elements.

Method Summary

Concrete Methods

Modifier and Type	Method	Description
static Collector.Characteristics	valueOf(String name)	Returns the enum constant of this type with the specified name.
static Collector.Characteristics[]	values()	Returns an array containing the constants of this enum type, in the order they are declared.

Methods inherited from class java.lang.Enum

clone, compareTo, equals, finalize, getDeclaringClass, hashCode, name, ordinal, toString, valueOf

Methods inherited from class java.lang.Object

getClass, notify, notifyAll, wait, wait, wait

Enum Constant Detail

CONCURRENT

```
public static final Collector.Characteristics CONCURRENT
```

Indicates that this collector is *concurrent,* meaning that the result container can support the accumulator function being called concurrently with the same result container from multiple threads.

If a CONCURRENT collector is not also UNORDERED, then it should only be evaluated concurrently if applied to an unordered data source.

UNORDERED

```
public static final Collector.Characteristics UNORDERED
```

Indicates that the collection operation does not commit to preserving the encounter order of input elements. (This might be true if the result container has no intrinsic order, such as a Set.)

IDENTITY_FINISH

```
public static final Collector.Characteristics IDENTITY_FINISH
```

Indicates that the finisher function is the identity function and can be elided. If set, it must be the case that an unchecked cast from A to R will succeed.

Method Detail

values

```
public static Collector.Characteristics[] values()
```

Returns an array containing the constants of this enum type, in the order they are declared. This method may be used to iterate over the constants as follows:

```
for (Collector.Characteristics c : Collector.Characteristics.values())
    System.out.println(c);
```

Returns:

an array containing the constants of this enum type, in the order they are declared

valueOf

```
public static Collector.Characteristics valueOf(String name)
```

Returns the enum constant of this type with the specified name. The string must match *exactly* an identifier used to declare an enum constant in this type. (Extraneous whitespace characters are not permitted.)

Parameters:

name - the name of the enum constant to be returned.

Returns:

the enum constant with the specified name

Throws:

IllegalArgumentException - if this enum type has no constant with the specified name

NullPointerException - if the argument is null

Submit a bug or feature
For further API reference and developer documentation, see Java SE Documentation. That documentation contains more detailed, developer-targeted descriptions, with conceptual overviews, definitions of terms, workarounds, and working code examples.
Copyright © 1993, 2026, Oracle and/or its affiliates. All rights reserved. Use is subject to license terms.
Also see the documentation redistribution policy.

Scripting on this page tracks web page traffic, but does not change the content in any way.